

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 2:

Alternative and choice

► What you will learn in this chapter

- A set of principles for understanding the economics of how individuals make choices
- Production possibility frontier and opportunity cost

CHOICE: THE HEART OF ECONOMICS.

Gemini

Claude



ChatGPT

Pi

INDIVIDUAL CHOICE: THE PRINCIPLES

1. Choices are necessary because resources are scarce.

RESOURCES AND SCARCITY

Resource: anything that can be used to produce something else.

Scarce: in short supply; a *resource* is scarce when there is not enough of the resource available to satisfy all the various ways a society wants to use it.

INDIVIDUAL CHOICE: THE PRINCIPLES

2. The true cost of something is its opportunity cost.

Opportunity cost: what you must give up in order to get something.



Tesla must decide whether to allocate its limited factory capacity to produce more car batteries or pivot to producing home energy storage systems (Powerwalls).

1.Incentives:

- **Profit Motive:** Car batteries generate higher margins per unit (15,000vs.15,000vs.10,000 for Powerwalls).
- **Strategic Incentive:** Powerwalls align with Tesla's mission to accelerate sustainable energy adoption.

2.Marginal Decision-Making:

- **Marginal Benefit of Car Batteries:**
 - \$15,000 profit per unit.
 - Strengthens EV market dominance.
- **Marginal Benefit of Powerwalls:**
 - \$10,000 profit per unit.
 - Diversifies revenue streams and supports energy grid stability.

3.Marginal Analysis:

- **Marginal Cost Comparison:**
 - Car batteries: \$12,000/unit (raw materials, labor).
 - Powerwalls: \$7,000/unit (simpler design).
- **Net Marginal Benefit:**
 - Car batteries: $15,000 - 15,000 - 12,000 = +\$3,000/\text{unit}$.
 - Powerwalls: $10,000 - 10,000 - 7,000 = +\$3,000/\text{unit}$.

4.Trade-offs & Opportunity Cost:

- **Trade-off 1:** Choosing car batteries means losing the chance to grow the energy storage market.
- **Trade-off 2:** Choosing Powerwalls risks ceding EV market share to competitors like BYD.
- **Opportunity Cost:** The profit (\$3,000/unit) and strategic advantage *lost* by not choosing the alternative.

INDIVIDUAL CHOICE: THE PRINCIPLES

3. “How much” is a decision at the margin.

Trade-off: comparison of the costs and the benefits of doing something.

Each bite of the candy bar has costs and benefits.



INDIVIDUAL CHOICE: THE PRINCIPLES

Marginal decision: decision made at the margin of an activity about whether to do a bit more or a bit less of that activity.

Marginal analysis: the study of marginal decisions.

Incentive: anything that offers rewards to people who change their behavior.

LEARN BY DOING: DISCUSS

Congestion pricing in New York City, also known as the Central Business District Tolling Program, began on January 5, 2025. It applies to most motor vehicular traffic using the central business district area of Manhattan south of 60th Street, known as the Congestion Relief Zone, in an effort to encourage commuters to use public transportation instead.

Discuss: WOULD YOU SUPPORT THIS PLAN? Why or why not?

INDIVIDUAL CHOICE: THE PRINCIPLES

4. People usually respond to incentives, exploiting opportunities to make themselves better off.



In the United States, restaurant customers have the option of adding a tip to the restaurant bill. In much of Europe a tip is added automatically.

Where would you expect waiters to be more attentive?

LEARN BY DOING: BUSINESS CASE

CAN YOU SPOT THE PROBLEM WITH THIS INCENTIVE SYSTEM?

A juvenile prison in Pennsylvania offered judges bribes the more young people they sent to jail.

One teen was jailed for a Facebook page that was critical of her high-school principal.

Both judges pleaded guilty to accepting \$2.6 million in bribes.



INTERACTION AND INDIVIDUAL CHOICE

5. There are gains from trade.



Trade allows us all to consume more than we otherwise could.



BEST IF USED BY AUG 22 23
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Nutrition Facts

40 servings per container
size 1 oz (28g/about 1/4 cup)

Amount per serving

Calories

170

% Daily Value*

Total Fat 16g	20%
Saturated Fat 2g	11%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 85mg	4%
Total Carbohydrate 6g	2%
Dietary Fiber 2g	6%
Total Sugars 1g	
Includes 0g Added Sugars	0%
Protein 5g	

Vitamin D 0mcg 0% • Calcium 40mg 2%
Iron 1.3mg 6% • Potas. 170mg 4%

*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

INGREDIENTS: Cashews, Almonds, Pecans, Brazil Nuts, Macadamia Nuts, Peanut Oil*, Sea Salt.

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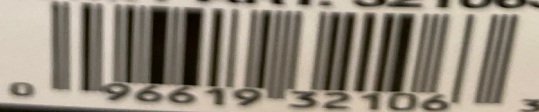
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INTERACTION AND INDIVIDUAL CHOICE



Specialization: the situation in which each person specializes in the task that he or she is good at performing.

1.Specialization:

1. Countries focus on producing what they're best at (e.g., U.S. tech, Saudi oil).
2. This increases efficiency and lowers costs.

2.Comparative Advantage:

1. Even if one country is better at producing everything, it's still better to trade.
2. Example: If the U.S. is better at making cars and computers than Mexico, but Mexico is relatively better at making cars, the U.S. should focus on computers and trade for cars.

3.More Choices for Consumers:

1. Trade gives people access to goods and services they can't produce domestically (e.g., tropical fruits, rare minerals).

4.Lower Prices:

1. Competition from imports keeps prices low (e.g., cheaper electronics from China).

5.Economic Growth:

1. Trade creates jobs in export industries and boosts overall economic activity.

1.Job Losses in Certain Industries:

1. Manufacturing jobs moved to countries with cheaper labor (e.g., China, Mexico).
2. Example: U.S. factories closing as companies relocate abroad.

2.Trade Deficits:

1. The U.S. imports more than it exports, leading to concerns about losing economic power.

3.National Security Concerns:

1. Reliance on other countries for critical goods (e.g., semiconductors, rare earth metals) is seen as risky.

4.Political Pressure:

1. Workers and industries hurt by trade lobby for protectionist policies (e.g., tariffs, quotas).

5.Unfair Trade Practices:

1. The U.S. accuses other countries (e.g., China) of unfair practices like subsidies, intellectual property theft, and dumping cheap goods.

6.Focus on Domestic Production:

1. Policies like “Buy American” aim to bring jobs back and reduce dependence on foreign goods.

INTERACTION AND INDIVIDUAL CHOICE

6. Markets move toward equilibrium.

Equilibrium: an economic situation in which no individual would be better off doing something different.



LEARN BY DOING: APPLICATION VIDEO

For a look at a real-world equilibrium, click the picture below to see the Aalsmeer Dutch Tulip auction in action. (4 minutes)



MODELS IN ECONOMICS

Model: a simplified representation of a real situation that is used to better understand real-life situations.



“OTHER THINGS EQUAL”

Other things equal assumption: all other relevant factors remain unchanged.

Ceteris paribu

We try to treat economics as close to a laboratory science as possible—with only one variable allowed to change at a time.



THE PRODUCTION POSSIBILITIES FRONTIER

**The PPF is
a diagram
that shows the
combinations of
two goods that are
possible for a society
to produce at full
employment.**

TRADE-OFFS: THE PPF

We can use the PPF model to answer questions like:

How much can we produce?

What will it cost us to change our mix of production?

Does it make sense to import the good from somewhere else?



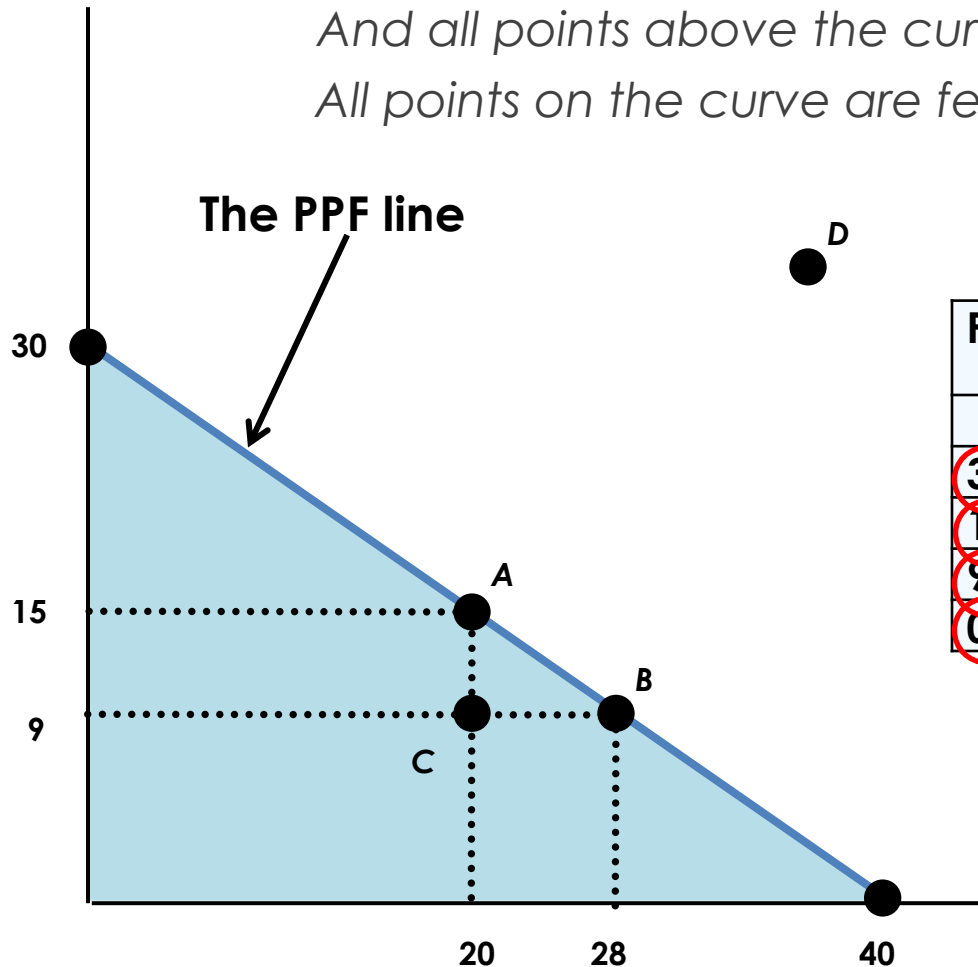
TRADE-OFFS: THE PPF

All points under the curve are feasible (but not efficient)...

And all points above the curve are not feasible.

All points on the curve are feasible and efficient.

The PPF line



Production Possibilities	
Dreamliners	Small Jets
30	0
15	20
9	28
0	40

OPPORTUNITY COST

Opportunity Cost: what must be given up in order to get a good.

Production Possibilities	
Dreamliners	Small Jets
30	0
15	20
9	28
0	40

EACH SMALL JET HAS AN OPPORTUNITY COST OF $\frac{3}{4}$ OF A DREAMLINER.

LEARN BY DOING: PRACTICE QUESTION

Suppose you are stranded on an island. Luckily, this island is rich in clams and mangos. If you devote all of your time to harvesting clams, you can get **100 clams in a week**. If you use all of your time to collect mangos, you can find **200 mangos in a week**.

Assume it is possible to collect fractional amounts of both goods and **draw a sketch of your production possibility frontier for a week**. (Place mangos on the x-axis and clams on the y-axis.)

Calculate the opportunity cost of each good.

LEARN BY DOING: PRACTICE QUESTION

Combination	Number of lamps possible	Number of sofas possible
<i>A</i>	0	1,000
<i>B</i>	50	800
<i>C</i>	125	500
<i>D</i>	200	200
<i>E</i>	225	100
<i>F</i>	250	0

What is the opportunity cost of each lamp?

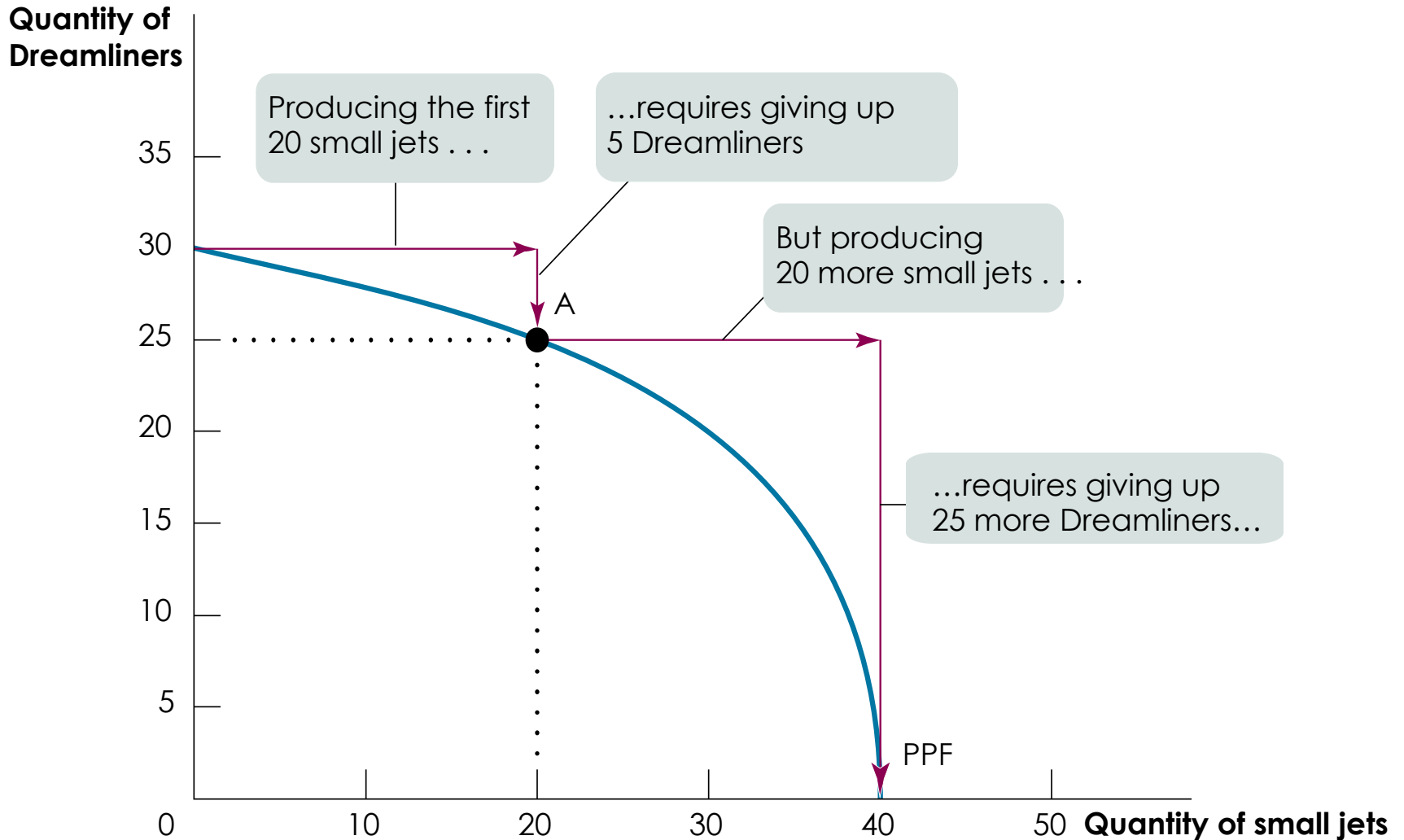
- a) $\frac{1}{4}$ of a sofa
- b) 1 sofa
- c) 4 sofas
- d) 200 sofas

LEARN BY DOING: PRACTICE QUESTION

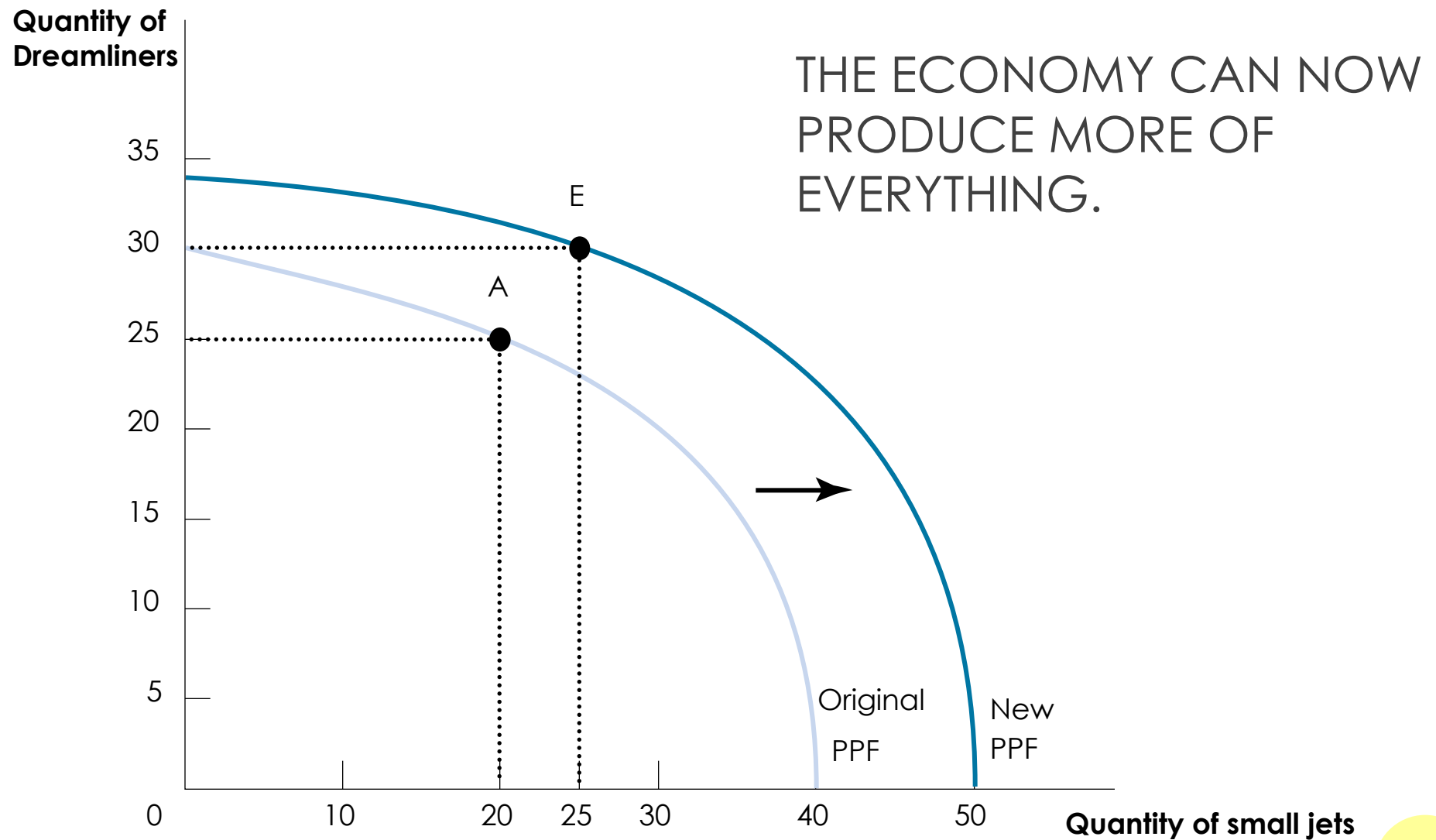
With a partner, create your production possibilities curve for the following two activities: (a) drawing happy faces and (b) drawing elephants:

1. Take turns timing each other for 30 seconds and recording the following:
 - The maximum number of happy faces drawn
 - The maximum number of elephants drawn
2. Now construct your PPF with elephants on the vertical axis and faces on the horizontal axis.
3. Calculate your opportunity cost for each good.

INCREASING OPPORTUNITY COST



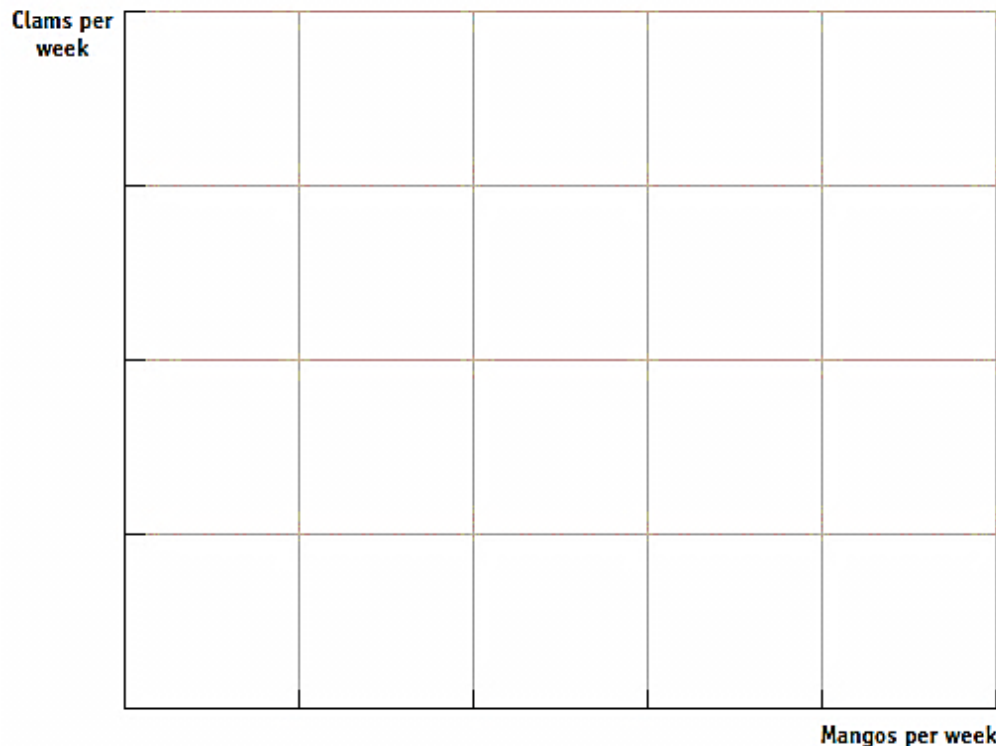
ECONOMIC GROWTH



LEARN BY DOING: PRACTICE QUESTION

If you invent a clam rake that doubles your clam production, what happens to your opportunity cost?

Graph the effect of your new invention on the original PPF.



WHAT CAUSES ECONOMIC GROWTH?

Two possibilities:

1. **An increase in factors of production:** resources used to produce goods and services.
2. **Better technology:** the technical means for producing goods and services.

FACTORS OF PRODUCTION

1. **Land** includes natural resources, such as mineral deposits, oil, natural gas, water, and actual land acreage.



FACTORS OF PRODUCTION



2. **Labor** is the mental and physical abilities of the workforce.

FACTORS OF PRODUCTION

3. ***Physical capital*** is manufactured items used to produce other goods and services.



ECONOMIC RESOURCES



4. ***Human capital*** is the educational achievements and skills of the labor force (which increase labor productivity).

LEARN BY DOING: APPLICATION VIDEO

Check out how containerization shaped the modern world. Click [here](#) or on the picture below. Consider how the world's PPF changed. (5 minutes)

